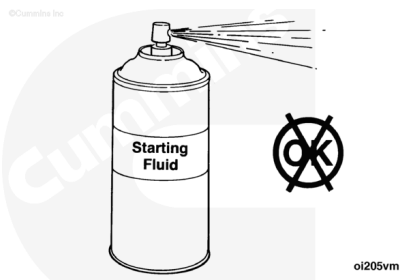


General Information

⚠ CAUTION ⚠

Do not, under any circumstance, apply an external heat source to the lubricating oil pan. Due to the composite materials used in the manufacture of the lubricating oil pan, heat applied directly or indirectly can damage the oil pan.



⚠ CAUTION ⚠

Due to the potential for engine damage, do not use starting fluid. The use of starting fluid has no practical purpose as a cold weather starting aid in a gaseous fueled engine. The use of starting fluid will cause engine damage.

It is possible to operate gas engines in extremely cold environments if they are properly prepared and maintained. The correct lubricants, fuels, and coolant **must** be used for the cold weather range for which the vehicle is being operated. See the chart below for recommendations in different operating ranges.

Winterize 0°C [32°F] to -23°C [-9°F]	Winterize -23°C [-9°F] to -32°C [-26°F]	Winterize -32°C [-26°F] to -54°C [-65°F]
Use ethylene glycol antifreeze to protect to -29°C [-20°F].	Use 50 percent ethylene glycol antifreeze 50 percent water mixture.	Use 60 percent ethylene glycol antifreeze 40 percent water mixture.
Use multi-viscosity oils meeting American Petroleum Institute (API) CD specifications for natural gas engines.	Use Arctic oil.	Use Arctic oil.

Cold Weather Operating Aids

Temperature	Coolant Heater	Oil Heater	Under-hood Air	Battery Heater	Radiator Shutters	Engine Enclosure	Thermostatic Fan
50 to 32°F 10 to 0° C							<i>Suggested</i>
32 to -10°F 0 to -23° C	-12°C [11°F]	-12°C [11°F]	-9°C [15°F]	-9°C [15°F]	-0°C [32°F]	-19°C [-2°F]	-19°C [-2°F]
-10 to -25°F -23 to -32° C	Required	Required*	Required	Required	Required	Required	Required
-25 to -65°F -32 to -54° C							

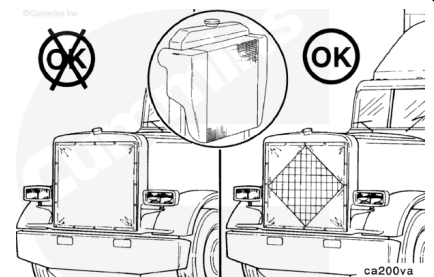
* Required dependent upon viscosity/pour point.

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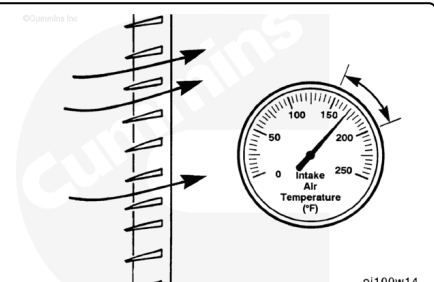
The cold weather operating aids are required for cold weather operation.

Winterfronts and Shutters

Winterfronts can be used on a vehicle equipped with a charge air cooler, but **must** be designed to partially cover the frontal area of the cooling system. A minimum of 120 square inches (11 in x 11 in) of frontal area **must** be left open to air flow for the charge air cooler to function correctly.



Installations of charge air cooler engines with shutters also requires an air intake manifold air temperature switch to open the shutters to prevent excessive intake manifold temperatures. This prevents engine damage due to high intake manifold temperatures as a result of blocked air flow across the charge air cooler.



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